Report 8

Routing and Inter-VLAN Routing

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NETWORK SECURITY – BATCH 2

Introduction:

In Week 8, I studied and implemented **static routing, dynamic routing, and inter-VLAN routing** using Cisco Packet Tracer. The main objective was to understand how routers forward traffic between different networks and how devices in separate VLANs can communicate through a Layer 3 device.

The practical implementation was performed on the existing Week 5 enterprise topology by adding 802.1Q trunking and Router-on-a-Stick inter-VLAN routing.

Static Routing:

Static routing is a routing method in which the administrator manually configures routes on a router. A static route specifies the destination network and the next-hop address or outgoing interface.

Example:

ip route 192.168.30.0 255.255.255.0 10.0.0.6

This tells the router where to forward traffic destined for the 192.168.30.0/24 network.

Advantages:

- Simple for small networks
- Provides direct administrative control
- Does not require a routing protocol
- Uses fewer network resources

Disadvantages:

- Routes must be configured manually
- Difficult to maintain in large networks
- Does not automatically adapt to topology changes

Real-World Example:

A small branch office with only a few networks can use static routes because there are relatively few routes to maintain.

Dynamic Routing:

Dynamic routing allows routers to automatically exchange routing information using routing protocols.

Common dynamic routing protocols include:

- OSPF
- EIGRP
- RIP
- BGP

For example, OSPF can automatically learn available networks and select suitable paths.

Advantages:

- Automatically learns routes
- Adapts to network changes
- Suitable for large networks
- Supports redundant paths

Disadvantages:

- Requires additional configuration
- Uses CPU, memory, and bandwidth
- More complex than static routing

Real-World Example:

Large enterprises with many branches and routers commonly use dynamic routing because manually configuring every route would be difficult.

Static vs Dynamic Routing:

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Automatic
Best for	Small networks	Medium/Large networks
Route changes	Manual	Automatic
Scalability	Low	High
Resource usage	Low	Higher
Common enterprise use	Small/simple networks	Dynamic routing such as OSPF

Existing VLAN Structure:

The Week 5 topology already contained departmental VLANs

Switch	Department	VLAN
SW1	SALES	10
SW2	HR	40
SW3	IT	20
SW4	FINANCE	30

Inter-VLAN Routing:

Devices in different VLANs cannot communicate directly through Layer 2 switching because each VLAN represents a separate broadcast domain. A Layer 3 device, such as a router or Layer 3 switch, is required to route traffic between the VLANs. In this implementation, Router 2 was used for inter-VLAN routing through Router-on-a-Stick.

Router-on-a-Stick:

Router-on-a-Stick allows a single physical router interface to route traffic between multiple VLANs by using subinterfaces. The router interface connected to the switch operates as a trunk, and each VLAN is assigned its own subinterface.

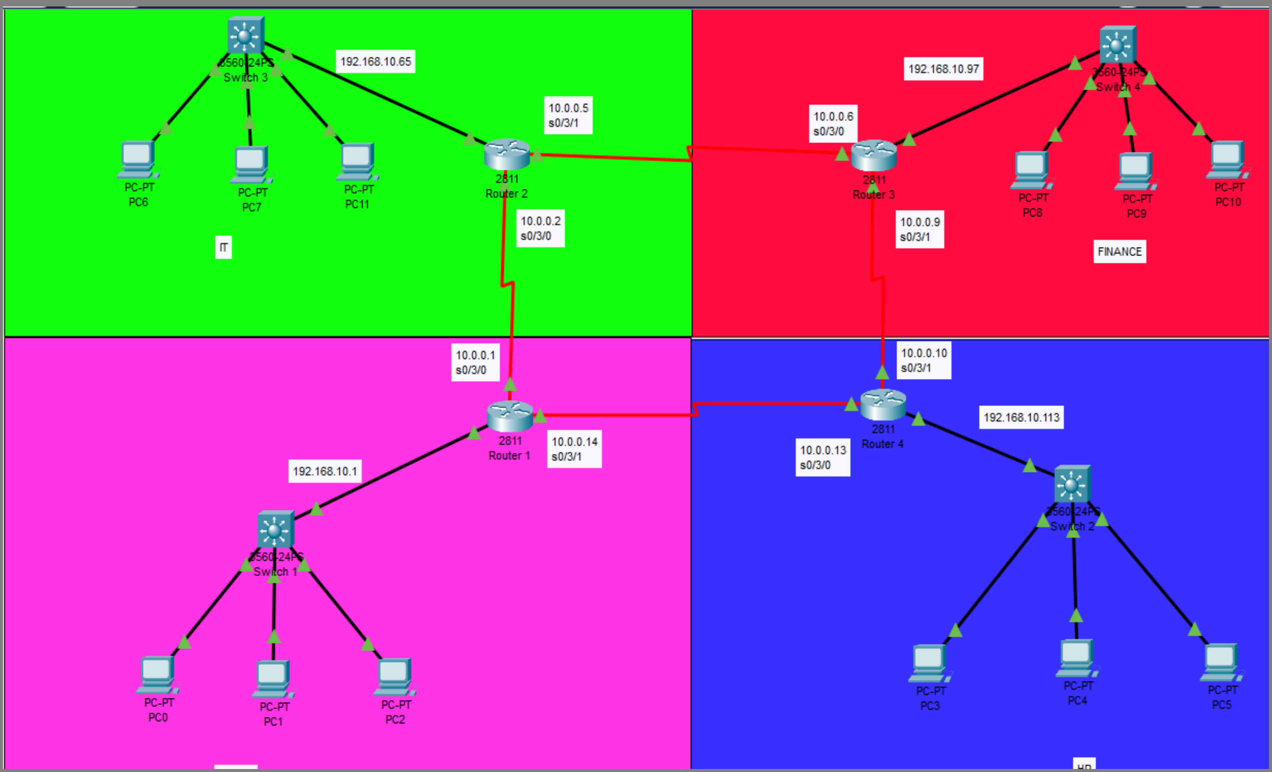
VLAN 20 and VLAN 30 Configuration:

For the practical inter-VLAN demonstration:

- VLAN 20 = IT
- VLAN 30 = FINANCE

The trunk between SW3 and R2 carries VLAN 20 and VLAN 30 traffic.

TOPOLOGY:



SW3 Configuration:

```
SW3#
SW3#
SW3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#int f0/4
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 30
SW3(config-if)#no shutdown
SW3(config-if)#
SW3(config-if)#
SW3(config-if)#
SW3(config-if)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

SW3#w
Building configuration...
[OK]
SW3#
```

VLAN 20 was already configured as the IT VLAN.

Configure the router connection as a trunk:

The connection between SW3 Fa0/1 and R2 Fa0/0 was configured as a trunk

```
SW3(config)#
SW3(config)#int f0/1
SW3(config-if)#switchport mode trunk

SW3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

SW3(config-if)#switchport trunk allowed vlan 20,30
SW3(config-if)#no shutdown
SW3(config-if)#exit
SW3(config)#exit
SW3#
%SYS-5-CONFIG_I: Configured from console by console
```

Configure the Finance PC port:

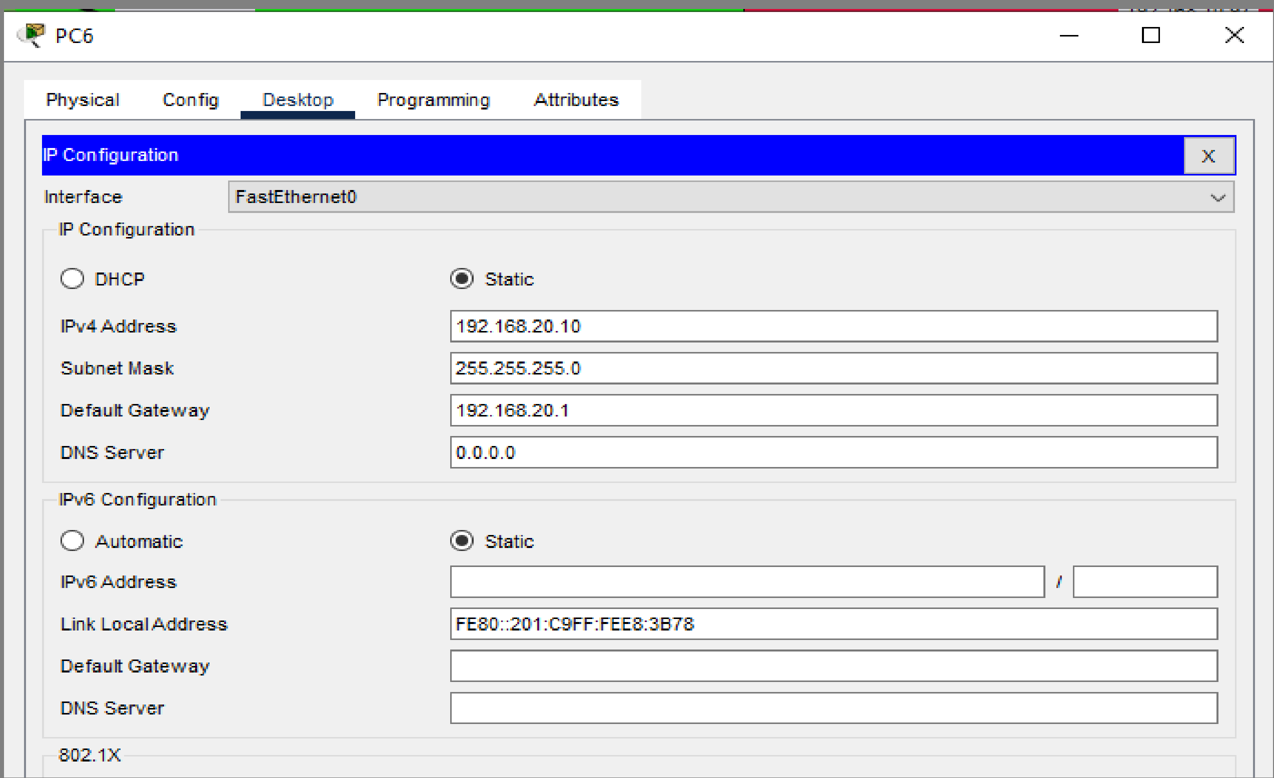
```
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#int f0/4
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 30
SW3(config-if)#no shutdown
SW3(config-if)#
SW3(config-if)#
SW3(config-if)#
SW3(config-if)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

SW3#w
Building configuration...
[OK]
SW3#
```

R2 Subinterface Configuration:

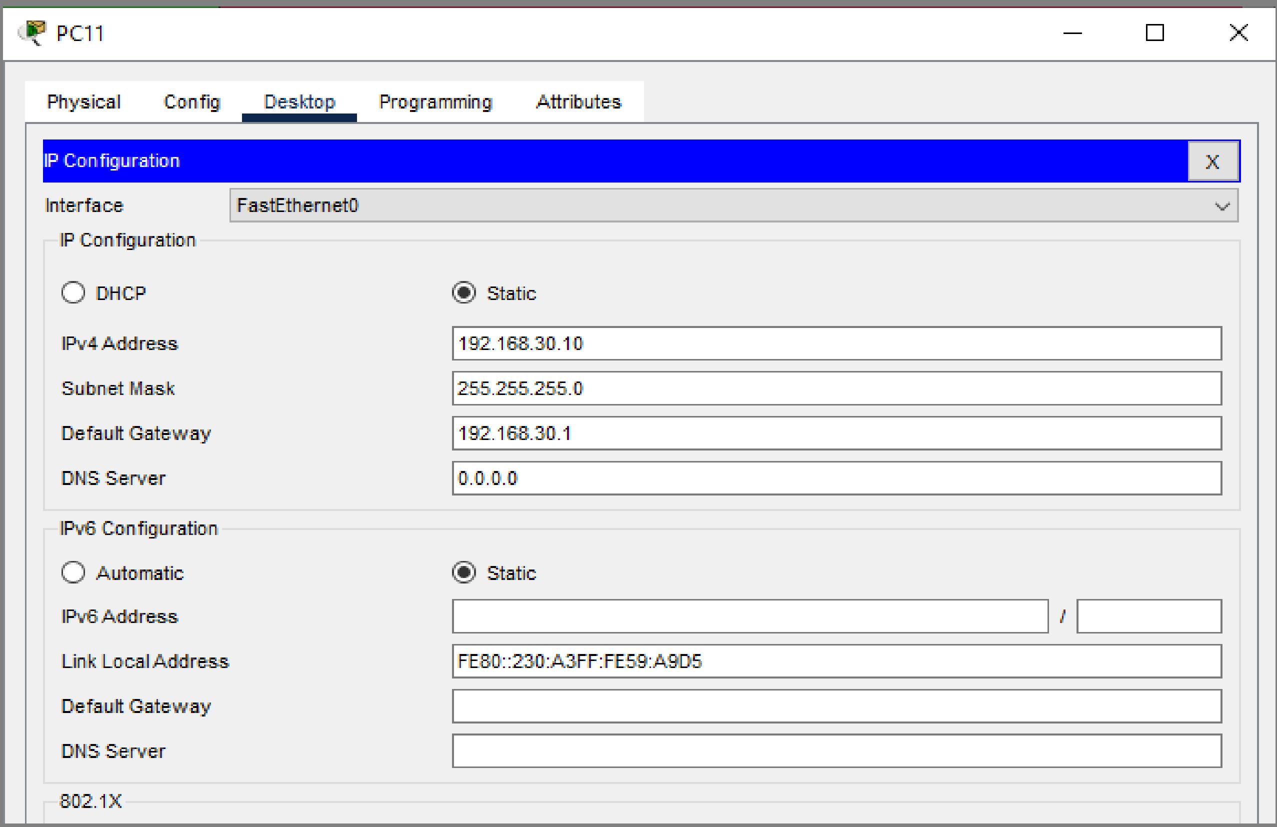
```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#no ip add
R2(config-if)#no shutdown
R2(config-if)#
R2(config-if)#
R2(config-if)#int f0/0.20
R2(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

R2(config-subif)#encapsulation dot1Q 20
R2(config-subif)#ip add 192.168.20.1 255.255.255.0
R2(config-subif)#exit
R2(config)#
R2(config)#int f0/0.30
R2(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to up

R2(config-subif)#encapsulation dot1Q 30
R2(config-subif)#ip add 192.168.30.1 255.255.255.0
R2(config-subif)#exit
R2(config)#
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

PC IP Configuration:

IT PC – VLAN 20



Finance PC – VLAN 30:



VLAN Verification:

```
SW3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
20   IT                               active    Fa0/2
30   GUEST                            active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW3#
```

This verified that the required VLANs were present and that the correct access ports were assigned.

Trunk Verification:

```
SW3#
SW3#
SW3#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       20,30

Port        Vlans allowed and active in management domain
Fa0/1       20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       20,30
```

Therefore, VLAN 20 and VLAN 30 traffic can travel between SW3 and R2

Router Interface Verification:

```
[OK]
R2#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES manual up                    up
FastEthernet0/0.20     192.168.20.1    YES manual up                    up
FastEthernet0/0.30     192.168.30.1    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.2        YES manual up                    up
Serial0/3/1            10.0.0.5        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Routing Table Verification:

```
R2>en
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
C       10.0.0.1/32 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
C       10.0.0.4/30 is directly connected, Serial0/3/1
L       10.0.0.5/32 is directly connected, Serial0/3/1
C       10.0.0.6/32 is directly connected, Serial0/3/1
     192.168.10.0/24 is variably subnetted, 3 subnets, 3 masks
S       192.168.10.0/26 [1/0] via 10.0.0.1
S       192.168.10.96/28 [1/0] via 10.0.0.6
S       192.168.10.112/29 [1/0] via 10.0.0.6
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks

R2#
```

Inter-VLAN Connectivity Test:

Connectivity was tested between the two VLANs. From the VLAN 20 PC:

ping 192.168.30.10

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

A successful response confirmed that traffic was routed from VLAN 20 to VLAN 30.

The reverse test was also performed:

ping 192.168.20.10 from the VLAN 30 PC

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Static and Dynamic Routing in Enterprise Networks

Static routing is suitable when the network is small and stable because routes are easy to configure manually.

Dynamic routing is preferred as networks become larger because routers can automatically learn routes and adapt when links or paths change.

For enterprise networks, dynamic routing protocols such as OSPF are commonly used because they improve scalability and simplify route management.

Challenges Faced

During the implementation, the main challenge was ensuring that the VLAN traffic was correctly carried between the switch and router. The trunk initially needed to be configured with the correct VLANs, and the correct access port had to be assigned to VLAN 30. After configuring the trunk, router subinterfaces, VLAN membership, and PC gateways, inter-VLAN communication was successfully established.

Learning Experience

This week improved my practical understanding of static routing, dynamic routing, VLAN segmentation, 802.1Q trunking, and inter-VLAN routing. I learned how Router-on-a-Stick uses router subinterfaces to route traffic between VLANs and how to verify the configuration using Cisco IOS commands. The practical testing also improved my troubleshooting skills and understanding of how Layer 2 and Layer 3 technologies work together.

Conclusion

Week 8 focused on routing and inter-VLAN communication. Static and dynamic routing were studied and compared, with dynamic routing such as OSPF being more suitable for larger enterprise networks. The existing VLAN topology was extended using 802.1Q trunking and Router-on-a-Stick. VLAN 20 and VLAN 30 were successfully configured on the same switching path, and Router 2 was configured with subinterfaces to route traffic between them. Successful ping testing confirmed that inter-VLAN routing was functioning correctly.